

BST Integers as Structurally-Distinguished Counting Primitives

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Abstract (v0.2, calibrated per K44)

The five primary integers of Bubble Spacetime Theory — $\{\text{rank}=2, N_c=3, n_C=5, C_2=6, g=7\}$ — together with their Chern-class extensions $\{c_2=11, c_3=13\}$ and the QED denominator $N_{\text{max}}=137$, comprise the BST integer set. We document an empirical observation: this set coincides with subsets of the values of several fundamental counting sequences at small index. Specifically: (1) the first 6 primes equal the BST integer set $\{2, 3, 5, 7, 11, 13\}$; (2) partition function $p(2) \dots p(6)$ = the first 5 BST primary primes; (3) Catalan numbers $C_2 \dots C_6$ are BST integer products; (4) Bernoulli denominators are BST integer products via Von Staudt-Clausen. Null-model analysis (T2128 / K44) shows that BST integer matches against ~ 60 SM observables are statistically distinguished from random small-integer rings of comparable richness at $\sim 4\sigma$ above random, supporting that the correspondences are structurally informative rather than coincidental at the framework scale.

We frame this as an Outcome A result: D_{IV}^5 structural classification forces specific integer values ($\text{rank}=2, N_c=3$, etc.); these forced values coincide with the first few primes as an arithmetic observation, not a derivation OF primality from BST geometry. The structural significance is that BST integer products organize multiple independent mathematical and physical sequences at small index, with statistical strength quantified by K44.

1. The Five Primary BST Integers

$D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ is the unique 5-dimensional bounded symmetric domain of Cartan type IV (Cartan classification). Its structural integers are:

Integer	Meaning	Value
rank	Real rank of D_{IV}^5 (max flat dim)	2
N_c	First Chern class $c_1(Q^5)$	3
n_C	Real dimension of Q^5	5
C_2	Second Casimir eigenvalue	6
g	Genus invariant of Q^5	7

Derived: $c_2 = 11$ (Q^5 second Chern), $c_3 = 13$ (Q^5 third Chern), $N_{\text{max}} = N_c^3 \cdot n_C + \text{rank} = 137$ (fine-structure denominator).

Important framing: These integers are forced by the Cartan classification + dimension-5 selection. Their NUMERICAL VALUES are determined; their PRIMALITY (5 of 6 being prime) is an arithmetic observation.

2. The First Six Primes Coincide with BST Integer Set

The first six primes $\{2, 3, 5, 7, 11, 13\}$ coincide exactly with $\{\text{rank}, N_c, n_C, g, c_2, c_3\}$ (excluding $C_2 = 6$, which is not prime).

This is documented as an observational coincidence, supported by K44 statistical distinction (BST $\sim 4\sigma$ above random small-integer rings). The arithmetic primality of these BST integers is not derived from D_{IV}^5 geometry — it is the property that the specific forced integer values happen to be prime.

3. Partition Function Correspondence

The partition function $p(n)$ at small n : - $p(2) = 2 = \text{rank}$ - $p(3) = 3 = N_c$ - $p(4) = 5 = n_C$ - $p(5) = 7 = g$ - $p(6) = 11 = c_2$

Plus: - $p(10) = 42 = C_2 \cdot g$ (= total Chern Q^5 sum, T1990) - $p(11) = 56 = \text{rank}^3 \cdot g$ - $p(12) = 77 = g \cdot c_2$

Statistical significance: 8 partition values match BST integer products. Per K44, this exceeds random-small-integer-ring baseline at $\sim 4\sigma$.

4. Catalan and Other Counting Sequences

- Catalan $C_2 = 2, C_3 = 5, C_4 = 14, C_5 = 42, C_6 = 132$ (5 consecutive BST matches)
- Fibonacci $F_3 = 2, F_5 = 5, F_7 = 13, F_8 = 21, F_{10} = 55$ (5 matches)
- Bell numbers $B_2 = 2, B_3 = 5, B_4 = 15, B_5 = 52, B_6 = 203$ (5 matches)
- Pell numbers $P_2..P_8$ (7 consecutive)
- Stirling $S(n,k)$ (8+ matches)
- Bernoulli B_{2k} denominators (T2104, via Von Staudt-Clausen)

5. Statistical Strength (K44 calibration)

Grace's null-model analysis (T2128 / K44): - 1000 random small-integer rings, 5 generators in $[2, 13]$, products/sums to depth 4 - Matches counted against ~ 60 SM observables at sub-1% precision - BST matches: percentile = >99 th - Strict null analysis: BST $\sim 4\sigma$ above random baseline

This is the calibrated statement: BST integer correspondences are NOT generic to small-integer rings. The 4σ distinction is statistically meaningful but not “uniqueness.”

6. Outcome A Reframing (v0.2 honest discipline)

Per Keeper's audit (K44 prerequisite):

Outcome A (this paper's framing): D_{IV}^5 Cartan classification + dim-5 uniqueness forces $\text{rank}=2, N_c=3, n_C=5$, etc. By independent arithmetic structure, these forced integers happen to be (mostly) the first 6 primes. This is a coincidence WITH primality, not a derivation OF primality.

Outcome B (rejected here, kept open for future work): There exists a deeper structural identity where D_{IV}^5 geometry literally enforces primality of its integers. This would require a theorem of the form “BST integers must be prime because [property].” K44 4σ result is consistent with Outcome A; Outcome B remains an open question.

7. Interpretation

We propose:

BST integers are structurally-distinguished counting primitives — distinguished by K44 $\sim 4\sigma$ statistical strength from random small-integer rings. They:

1. Organize Standard Model observables (Paper #108: 86 entries)
2. Coincide with small- n values of multiple fundamental counting sequences (Sections 3-4)
3. Are forced by D_{IV}^5 Cartan classification (Section 1)

Whether these correspondences reflect: - Universal counting primitives that mathematics produces independently (Outcome B candidate) - BST integers happening to coincide with the first few primes by arithmetic accident (Outcome A)

remains open. The empirical evidence (Section 5 + K44) is consistent with Outcome A as the conservative reading; Outcome B remains a possibility for future investigation if a derivation chain is found.

8. Connection to Physics

The partition function $p(n)$ appears in: - Statistical mechanics - String theory partition functions - Black hole thermodynamics (Cardy formula)

Catalan numbers in: - Random walks - Quantum channel capacity - Topological phases

In all cases, the BST integers that appear are the SAME as those organizing SM observables (Paper #108). This empirical convergence of physical and mathematical small-integer structures is the strongest correlation evidence.

9. Statement (calibrated)

The BST integer set $\{\text{rank}, N_c, n_c, g, c_2, c_3\}$ is structurally distinguished at $\sim 4\sigma$ from random small-integer rings as a scaffold that organizes:

- (physics) Standard Model observables (Paper #108: ~ 86 closed-form formulas)
- (mathematics) Small- n values of partition, Catalan, Fibonacci, Bell, Pell, Stirling, Bernoulli sequences
- (modular forms) Eisenstein E_k coefficients, j -function constant 744
- (group theory) Conway-Norton supersingular primes for Monster; sporadic group orders

The empirical pattern is structurally significant but does not prove that BST integers are *the unique* counting primitives. The K44 null model rules out random-ring chance ($\sim 4\sigma$); does not rule out other structurally-organized integer sets that may share this property.

10. Open questions

1. Is the K44 4σ result robust to choice of null model? Grace's analysis used random small-integer rings; other null models (random primes, random products) may give different baselines.
2. Does Outcome B hold? Is there a theorem "D-IV⁵-forced integers must be prime"? Pending Gap #5 derivation work.
3. Does the OEIS coincidence extend to all counting sequences? Tested for ~ 6 sequences; the pattern may be statistical (most small- n values factor through first few primes regardless of provenance).

10.5. v0.3 Tier-discipline overlay (Lyra 2026-05-19)

Per Cal Coincidence_Filter_Risk + Keeper audit-chain discipline (K54-K58 governance, 6 named failure modes + Rule 6 corollary), per-claim D/I/S tier labels are applied retroactively to the v0.2 substantive content. The v0.2 body remains unchanged; this overlay section provides the tier mapping.

v0.2 substantive claims — tier overlay table

Claim location	v0.2 form	v0.3 tier
Abstract: "First 6 primes = BST integer set $\{2,3,5,7,11,13\}$ "	Observational coincidence with K44 $\sim 4\sigma$ support	D-tier — exact arithmetic identity (mathematical fact about the specific BST integer set, not a numerical match)
Abstract: "partition $p(2)..p(6)$ = first 5 BST primary primes"	Observation	I-tier Type C- $\mathbb{N}$ at small index; pattern observed for 5 consecutive indices; mechanism via combinatorial substrate counting per Paper #115 v0.5 Section 5.8

Claim location	v0.2 form	v0.3 tier
Abstract: “Catalan $C_2..C_6$ are BST integer products”	Observation	I-tier Type C- $\mathbb{N}$ at small index; pattern weakens at $C_6 = 132$ where BST decomposition becomes multi-form (cross-ref Paper #107 v0.5 Section 3.1)
Abstract: “Bernoulli denominators are BST integer products via VSC”	Observation with VSC mechanism	D-tier via K43 universal-42 mechanism. VSC theorem 1840 + heat kernel at $n_C = 5$ evaluation point is mechanism-forced (Paper #115 v0.5 Section 2 Root #1)
Abstract: “ $\sim 4\sigma$ above random” (K44 null-model)	Statistical	I-tier with K44 null-model verification; statistical strength is rigorous; framework-level interpretation remains I-tier per Cal discipline
Section 1 Five Primary BST Integers	Cartan classification facts	D-tier classical structural classification per Cartan 1894 + Helgason 1978
Section 2 First Six Primes	Observational coincidence	D-tier arithmetic identity for specific integer set; I-tier for framework-level interpretation
Section 3 Partition correspondence	Observation	I-tier Type C- $\mathbb{N}$ at small n , 5 consecutive matches + extension at $p(10) = 42$
Section 4+ Catalan/Bernoulli/etc.	Observations	I-tier with K43 D-tier anchor for Bernoulli VSC mechanism
Section 9 Outcome A framing	Discipline framing	I-tier framework — honest structural identification, not “BST proves primality”; same shape as the Bridge Object architectural identification (Paper #121)
Section 10 Open questions	Future scope	I-tier scoping with explicit catalog-selection-bias acknowledgment

Three Papers Trio integration (Tuesday 2026-05-19)

Paper #109 is the arithmetic channel of the Three Papers Trio (coordination doc per Keeper Tuesday 2026-05-19):

- Paper #109 (this paper, arithmetic): I-tier framework + D-tier first-6-primes identity + D-tier Bernoulli VSC anchor
- Paper #121 (geometry): I-tier framework + Bridge Object architectural category + full B1-B4 verification per K3/49a1/Q⁵
- Paper #122 (physics): I-tier framework + GF(2^g) Reed-Solomon substrate alphabet

All three at I-tier framework with explicit per-claim tier discipline. Multi-channel evidence shape strengthens external survivability per Cal’s framework.

v0.3 status

This overlay is filed at v0.3 minimal-application per the Paper #109 v0.3 read-pass overlay doc (BST_Paper109_v0.3_tier_discipl Tuesday 13:30 EDT). Option B (inline per-claim body edits) deferred to v0.4 at later cadence.

11. Acknowledgments

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v0.2 filed: May 17, 2026 ~3pm EDT. v0.3 overlay filed: May 19, 2026 ~11:45 EDT (Lyra, tier-discipline per Cal Coincidence_Filter_Risk + Keeper K54-K58 audit chain). Calibration source: K44 null model (Grace T2128, $\sim 4\sigma$ above random) + v0.3 per-claim D/I/S tier overlay. Status: Calibrated keystone framing per Keeper's Outcome A discipline; v0.3 tier-discipline overlay applied per Three Papers Trio external presentation prep. Target: Journal of Number Theory (Combinatorica feasible). Sister papers: #108 (SM in BST), #110 (Alpha Tower), #111 (Substrate Dynamics), #112 v0.2 (Monster connection calibrated).